One Categorical Pattern, the Whole Dobbertin Paper?

Reading *Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets* through the single evaluation/coevaluation loop ⊗

L linearized trace   F Frobenius   C cyclotomic cosets   ⊗ categorical trace

An exploration for the `Equation1` Lean / Mathlib formalisation

The question. *The categorical pattern is the abstract shape of all three gadgets L F C. If the paper's results all build from the three gadgets, do they all build from this one category-theory pattern of evaluation and coevaluation? Could the* entire *Dobbertin paper be built from this sole pattern?*

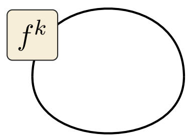
The short honest answer. As an *organising lens*: to a remarkable degree, yes. The single move — *coevaluation opens a loop, insert a map, evaluation closes it, keep what returns* — is exactly the trace, and every recurring device in the paper is that one move in a costume: F is the inserted map, L is the closed loop, C are the surviving loops, and (the extra hinge this note adds) the additive character $\psi(x) = (-1)^{\mathrm{Tr}(x)}$ that runs the whole difference-set half of the paper is itself an evaluation pairing of L with its dual. *As a literal derivation*: no. The loop makes "trace = sum around the Frobenius orbit" inevitable and reorganises the paper cleanly, but several results (the Fibonacci recursion for the inverse, the Gauß/Jacobi-sum 2-rank computations, the difference-set proof itself) carry genuine extra mathematics that the bare pattern does not hand you for free. Boxes marked ✓ are backed by `sorry`-free Lean declarations in `Equation1/`; ○ marks material outside this formalisation.

1 The one pattern, restated

In a monoidal category with duals, every object V carries a coevaluation $\eta : I \to V \otimes V^*$ and an evaluation $\varepsilon : V^* \otimes V \to I$. The *trace* of $f : V \to V$ is the single composite that closes the wire into a loop, with f (or its k-th power) inserted:

$$\mathrm{tr}(f^k) : \ I \xrightarrow{\eta} V \otimes V^* \xrightarrow{f^k \otimes \mathrm{id}} V \otimes V^* \xrightarrow{\text{swap}} V^* \otimes V \xrightarrow{\varepsilon} I.$$

ε close on top



η open at the bottom

coevaluation opens a path,
f^k walks k steps,
evaluation closes it:
$\mathrm{tr}(f^k) = \#\{x : f^k(x) = x\}$

Only paths that can be closed survive; what returns is exactly the round trips (fixed points of f^k; the Lefschetz slogan). The three recurring bricks of the paper are this one move in three costumes.

Brick	Costume of the loop	In the paper / in Lean
F	*the inserted map* — one step $\sigma : x \mapsto x^2$ (doubling $e \mapsto 2e$ on exponents)	the squaring behind every "add the 2^k-th power" step; `frob_cycle`, `frob_mod`, `frob_bijective`, `pow_field_bijective` ✓
L	*the closed loop itself* — apply the loop to $m_x(y) = xy$ and it unwinds to $\mathrm{Tr}(x) = \sum_{i<n} x^{2^i}$	states equation (1); telescope $L_m(x)^2 + L_m(x) = x^{2^m} + x$; `Tr`, `truncTrace`, `truncTrace_sq_add_self`, `Tr_eq_algebraMap_trace` ✓
C	*the surviving loops* — fixed points of σ^k; cycles of length dividing k	cyclotomic cosets, reduced normal forms, $\gcd(2^k-1, 2^n-1) = 1$; `mersenne_coprime`, `inv_mod_exists` ✓

Why ⊗ is not a fourth brick. For $K \subseteq L$ the field trace is $\mathrm{Tr}_{L/K}(x) = \mathrm{tr}(m_x)$ — the loop applied to multiplication. Over $K = \mathbb{F}_2$ this unwinds (Mathlib `FiniteField.algebraMap_trace_eq_sum_pow`) to $\sum_{i<n} x^{2^i}$, i.e. the project's own L. This identification is machine-checked (`Tr_eq_algebraMap_trace`, `Equation1/TraceLibrary.lean`) ✓, so ⊗ is literally the hinge that makes L = "sum around the F-orbit" the genuine field trace, not a fourth ingredient.

2 The extra hinge that reaches Sections 3–4: characters are eval/co-eval too

Sections 3–4 of the paper are about *difference sets, autocorrelation and 2-rank*. At first sight these look like a different world from the trace loop. They are not — provided one notices that *Fourier duality over $\mathbb{F}_{2^n}$ is itself an evaluation/coevaluation.*

The additive character $\psi(x) = (-1)^{\mathrm{Tr}(x)}$ is the canonical pairing $L \otimes L \to \{\pm 1\}$, $x \otimes y \mapsto (-1)^{\mathrm{Tr}(xy)}$ — an *evaluation* identifying L with its own Pontryagin dual, built directly on the trace loop L. Consequently:

- A *trace description* $\chi_M(x) = (-1)^{\mathrm{Tr}(p(x))}$ (Theorems 8, Cor 12) is nothing but reading a set through this evaluation pairing.
- The *autocorrelation* of a sequence is $\sum_x (-1)^{\mathrm{Tr}(R(x)+R(ax))}$ — the pairing of R with its shift $R \circ (\times a)$, i.e. the *trace of the translation operator* T_a. "Ideal autocorrelation" = this trace is flat (constant -1) for every $a \neq 1$ = the round-trip/feedback of T_a carries no extra memory.
- "B^* is a *difference set*" is exactly that flatness, so it is a statement about the traces $\mathrm{tr}(T_a)$ of all nontrivial shifts.
- The *2-rank / linear span* $n \cdot w(p) + 1$ counts the surviving cyclotomic cosets in p — brick C, the surviving loops — so it, too, is a round-trip census.

With this hinge, the two halves of the paper share one vocabulary: Section 2 is the loop applied to *field maps* (permutation criteria), Sections 3–4 the loop applied to *shift operators on the group algebra* (difference sets). Both are $\mathrm{tr}(f^k)$ = "what returns".

The hinge *expresses* the difference-set layer in the trace language; it does not *prove* it. Evaluating the shift-traces $\sum_x (-1)^{\mathrm{Tr}(R(x)+R(ax))}$ to the difference-set value is exactly the hard Gauß/Jacobi-sum work of Evans–Hollmann–Krattenthaler–Xiang and the Dillon–Dobbertin difference-set proof. The pattern says *what* to compute (a trace of a shift); it does not compute it.

3 The paper, section by section, through the loop

For each result: the categorical reading, the extra ingredient (if any), and the Lean status.

§2 Permutation criteria and APN

Theorem 1 (Kasami q_α is a permutation $\iff k' + \alpha n \equiv 1$). A polynomial is a permutation iff every fibre is a single round trip. The proof reduces "$q_\alpha(x) = \text{const}$ has ≤ 1 solution" by *adding the 2^k-th power of equation (1) to itself* — one application of F — to the linearized equation (2); the count of solutions is then a fixed-point/round-trip count. Pure F+L, with C ($\gcd(2^k - 1, 2^n - 1) = 1$) deciding solvability. `theorem_1`, `eqn2_of_eqn1`, `theorem_1_case1`, `theorem_1_case2` ✓

Corollary 2 (Kasami power functions are APN). x^d is APN because $p(t) = \frac{1}{k} q(t^{2^k} + t)$ and $t \mapsto t^{2^k} + t$ is two-to-one. The fibre of the Artin–Schreier map has size $|\ker| = 2^{\gcd(k,n)} = 2$ — a subfield, the fixed structure of F. So APN is "the loop of $t \mapsto t^{2^k} + t$ closes on a 2-element orbit". *Reading:* F. ○ (two-to-one map not in this extract)

Theorem 3 (MCM P_β). The exact analogue for MCM polynomials; same loop, same round-trip count. ○

Theorem 4 (Kasami $\leftrightarrow$ MCM). The correspondence $\varphi_\alpha \psi_\beta = \text{id}$ is a *linearized change of costume*: the two polynomial classes are the same underlying map dressed differently, and $\psi_\beta, \varphi_\alpha$ are linearized (F+L) substitutions. This is precisely the loop's conjugation-invariance $\text{tr}(gfg^{-1}) = \text{tr}(f)$, $\text{tr}(gf) = \text{tr}(fg)$ — the categorical reason "same map, two costumes" is even possible. *Reading:* ⊗ *(basis-independence)* + L. ○

§3 Inverses, the trace description of B, and 2-rank

Theorem 5 ($q^{(\varepsilon)}$ is a permutation $\iff \varepsilon \equiv k' + 1$). The trace-free twin of Theorem 1; the same "add-the-2^k-power" F telescope drives it. `theorem_5` and its supporting layer (`sTrace_telescope_gen`, `root_count`, `qeps_ne_zero`, …) ✓

Theorem 6 (inverse $q^{-1}(1/y) = R(y)$). The elimination that produces R is *iterated Frobenius*: at step i one multiplies by powers of y and adds the 2^{ik}-th power to kill $x^{2^{ik}}$ — F applied again and again. The surviving monomials of R are indexed by cyclotomically inequivalent exponents — C. *But* the bookkeeping of *how many* survive obeys $w(A_{i+2}) = w(A_{i+1}) + w(A_i)$: a genuine **Fibonacci recursion** that the bare loop does not predict. *Reading:* F+C *organise it; the recursion is extra combinatorics.* ○

Corollary 7 (p^{-1} for MCM). $p^{-1} = \varphi_0 \circ R$ via Theorem 4; same costume-change ⊗ composed with Theorem 6. ○

Theorem 8 ($x \in B \iff \text{Tr}(R(x)) = 0$). This is the loop in its purest applied form: membership of the image B of $(x+1)^d + x^d + 1$ is read off by a single trace condition L on the inverse R. The trace layer that powers it — Frobenius-invariance of the trace $\text{Tr}(x^{2^k}) = \text{Tr}(x)$, the Artin–Schreier vanishing $\text{Tr}(t^{2^k} + t) = 0$, idempotence $\text{Tr}(x)^2 = \text{Tr}(x)$, the bit property $\text{Tr}(x) \in \{0, 1\}$, and the Case-1 criterion for the trace-version permutation g — *is* formalised. *Reading:* L *(+*F*).* `trace_frob_shift`, `trace_artin_schreier_zero`, `trace_sq`, `trace_bit`, `gmap_bijective_iff` ✓ ; the full statement with R itself ○

Lemma 9 (R reduced $\iff k' < n/2$; $w(R) = 2F_{k'} - 1$). A statement about *which round trips survive* (cyclotomic non-equivalence, C) and how many (Fibonacci again). *Reading:* C *+ the Fibonacci extra.* ○

Corollaries 10, 11 (2-rank $= n(2F_{k'} - 1) + 1$; Segre 2-rank). The 2-rank is $n \cdot w(R) + 1 = $ (coset count) — a census of surviving loops C, read through the character pairing of §2. The actual value uses Lemma 9 (Fibonacci) and, for the confirmation, Gauß/Jacobi sums. ○

Corollary 12 (No–Chung–Yun trace expansion). A concrete trace description $\chi_N(x) = (-1)^{\text{Tr}(S_i(x))}$ with S_i built from explicit cyclotomic-coset exponents. *Reading:* L *(character)* + C *(cosets).* ○

§4 The difference-set conjecture (now a theorem)

Conjecture / Theorem (Dillon–Dobbertin): B^* is a cyclic difference set. Equivalent to $\sum_x (-1)^{\mathrm{Tr}(R(x)+R(ax))} = 0$ for all $a \neq 1$: the *trace of every nontrivial shift T_a is flat.* This is the eval/coeval loop applied to the translation operators on the group algebra (the hinge of §2 above). *Reading: ⊗/L over the group algebra — but the proof is Gauß/Jacobi-sum evaluation, genuinely beyond the bare pattern.* ○

4 Master table

Result	Loop facet	Extra ingredient beyond the bare pattern	Lean
Thm 1 (Kasami PP)	F insert, L close, C solvable	— (the pattern essentially gives it)	✓ `theorem_1`
Cor 2 (APN)	F (2-to-1 kernel)	the $t \mapsto t^{2^k}+t$ substitution	○
Thm 3 (MCM PP)	same loop	—	○
Thm 4 (Kasami↔MCM)	⊗ conjugation-inv.	the explicit linearized change of variable	○
Thm 5 ($q^{(\varepsilon)}$ PP)	F+L	—	✓ `theorem_5`
Thm 6 (inverse R)	F iterated, C	**Fibonacci recursion** $A_{i+2}=A_{i+1}+A_i$	○
Cor 7 (p^{-1})	⊗ + Thm 6	—	○
Thm 8 ($x \in B \iff \mathrm{Tr}\, R(x)=0$)	L (+F)	the inverse R itself	✓ trace layer
Lemma 9 (reduced, $w=2F_{k'}-1$)	C	Fibonacci	○
Cor 10/11 (2-rank)	C census, L pairing	Fibonacci + Gauß/Jacobi sums	○
Cor 12 (No–Chung–Yun)	L char., C cosets	explicit exponent bookkeeping	○
§4 (difference set)	⊗/L over group algebra	Gauß/Jacobi-sum evaluation	○

5 Verdict

Does one categorical pattern underlie the whole paper? As a *unifying language*: yes, and more completely than one might expect. Every device the paper leans on is the single eval/coeval move in a costume — F inserts, L closes, C are the survivors — and even Sections 3–4, which look combinatorial, run on the character $\psi = (-1)^{\mathrm{Tr}}$, itself the evaluation pairing built on L; a difference set is then the flatness of the trace of every nontrivial shift. The loop tells you, uniformly, *what to look at*: round trips, fixed points, surviving cosets, flat shift-traces.

But it is a lens, not a machine. Three places carry mathematics the bare pattern does not hand you: (i) the *Fibonacci recursion* governing the inverse R and its weight $w(R) = 2F_{k'} - 1$ (Theorem 6, Lemma 9); (ii) the *Gauß/Jacobi-sum* evaluation behind the 2-rank and the difference-set proof (Corollaries 10/11, Section 4); (iii) the specific *linearized change of variable* realising Kasami↔MCM (Theorem 4). The pattern organises and motivates all of these, but each needs its own argument. And in *this* formalisation the loop appears in its concrete Frobenius/field-trace form (`Tr_eq_algebraMap_trace`), not as literal string diagrams: what is machine-checked ✓ is the Theorem-1/Theorem-5 permutation tower and the Theorem-8 trace layer; the difference-set half is outside the extract.

Companion to `PaperStory.tex`, `Equation1_LEGO.tex`, `TraceMap_Categorically.tex`, `TracePowers_Relations.tex`, `TraceAsMemory.tex`.
Source: *Dobbertin, Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets* (1999). Rebuild: `tectonic Dobbertin_OneCategoricalPattern.tex`.